

Asad Ikram

Application for Doctoral student in Feminist AI Addressing Gender-Based Violence, KTH (PA-2026-2872)

Representative work

I have no peer-reviewed publications. The two documents below are my representative research work. Both are long, so as the advertisement asks, each is summarised here by its abstract. The full texts are uploaded as separate files with this application, because neither is yet available at a public web address.

1. Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment

Ikram, A. and Haig, E. · Manuscript, complete; not published and not submitted · Target venue: ICWSM 2027 · Asad Ikram is first author; the work develops his MSc dissertation, which Dr Haig supervised · Full text:

[Ikram_Haig_Is_the_Platform_Part_of_the_Measurement.pdf](#)

Comparing machine-labelled sentiment across social media platforms is now routine: differences in classifier scores between platforms are read as differences in public opinion. These comparisons rest on an assumption that the classifier measures the same construct, on the same scale, on every platform. To our knowledge, no study has tested that assumption for machine-labelled constructs by asking whether the platform is part of the measurement. We adapt the psychometric equivalence toolkit (generalizability theory, measurement invariance and differential item functioning, and score linking) to machine-labelled policy sentiment, treating the platform as a survey mode, and specify a six-step protocol with anchor events fixed in advance, an anchor-density audit, and a two-classifier convergence control. A simulation with known ground truth quantifies what the assumption costs in one generating configuration fixed in advance: across 500 replications, a pooled index that takes platform scores at face value misstates a known event shift by roughly an eighth, and because the platforms measure through different functions it reports the wrong sign of aggregate sentiment on nearly a quarter of days. A measurement-adjusted index recovers the shift with negligible bias and the correct sign on all but one day in sixty on average. A case study of the instrument from the first author's MSc dissertation, a fine-tuned BERT classifier over 255,446 UK economic-policy comments from Twitter/X, Reddit, YouTube, and Quora, motivates the design and illustrates the instrument-dependence the protocol addresses. The full four-platform application, under fresh collection with the pre-specified design, is the designated follow-up study. The protocol lets a researcher test whether pooling platforms is defensible in a given corpus, or whether platform scores must be linked before any average is taken.

2. Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies

Ikram, A. · MSc Data Analytics dissertation, University of Portsmouth, 2025, supervised by Dr Ella Haig · Project mark 78/100, degree awarded with Distinction · Sole author · Full text: [Asad_Ikram_MSc_Thesis.pdf](#)

Public sentiment on economic policies often first appears on social media, spread across platforms such as Twitter (X), Reddit, YouTube, and Quora. This dissertation presents a cross-platform sentiment analysis framework designed to unify and forecast citizen reactions to UK economic policies. Approximately 279,000 public social-media posts and comments related to taxation, welfare, inflation, and stimulus measures were collected using official APIs and licensed feeds. Posts were cleaned and labeled using the VADER sentiment analysis tool, and further analyzed using Natural Language Processing (NLP) and other techniques. A transformer-based model (BERT) and a Bidirectional LSTM were trained to classify sentiment (positive, neutral, negative), achieving high accuracy (BERT macro-F1 ≈ 0.88) on a validated set. Sentiment trends were aggregated over time, and Facebook Prophet was applied to forecast short-term public mood. The results reveal notable sentiment differences across policy domains, while the forecast model captured seasonal patterns and reactions around major policy announcements. Ethical and GDPR-compliant data handling was ensured, graphs were developed for policy analysts. The findings demonstrate that cross-platform NLP can effectively gauge public opinion in real time, providing policymakers with a proactive tool to understand and anticipate public response.

Both abstracts are reproduced unchanged from the original documents.